

Verifying kernel support

Make sure you have loaded the `nf_conntrack`, `nf_conntrack_ipv4` (and if your set-up also supports IPv6 and `nf_conntrack_ipv6`) and `nf_conntrack_netlink` modules. The simple way to verify if your kernel supports our requirements is to run a `sysctl -a` command, and `grep` out `conntrack`; and you should see the following output. If not, please consider compiling a more recent kernel version, or use an OS version which already fulfils the requirement.

```
[root@firewall1 libnetfilter_conntrack-0.9.10]# sysctl -a | grep -i
conntrack
net.netfilter.nf_conntrack_generic_timeout = 600
net.netfilter.nf_conntrack_acct = 1
net.netfilter.nf_conntrack_events = 1
net.netfilter.nf_conntrack_events_retry_timeout = 15
net.netfilter.nf_conntrack_max = 56432
net.netfilter.nf_conntrack_count = 3
net.netfilter.nf_conntrack_buckets = 13312
net.netfilter.nf_conntrack_checksum = 1
net.netfilter.nf_conntrack_log_invalid = 0
net.netfilter.nf_conntrack_expect_max = 200
net.nf_conntrack_max = 56432
```

State table synchronisation

In order to get `conntrackd` working in synchronisation mode, you have to fulfil the following requirements:

1. Have a high-availability manager (in our case, *keepalived*) in place. It takes care of the virtual IPs of the firewall cluster, detects errors and chooses when to migrate the virtual IPs from one cluster member to the other. However, only a virtual IP fail-over would not be sufficient for our needs; unless we have `conntrackd` installed, stateful fail-over will not work appropriately.
2. If you use a different high-availability manager, make sure it works correctly before going ahead.
3. You need a dedicated link between the firewalls, which will be used to transmit and receive the state information. This is mandatory for security reasons, to prevent anyone picking up the state information that is transferred between the firewalls. In our case, it would be on the *eth1* interface.
4. A properly configured stateful rule-set—otherwise you might experience problems during the fail-over. Hence, you need to maintain extra caution while designing the rule-set. I have shown a sample rule-set in a later part of this article.

To compile and install `conntrack-tools`, run the following commands:

```
tar -jxvf conntrack-tools-0.9.14.tar.bz2
cd conntrack-tools-0.9.14
[root@firewall1 conntrack-tools-0.9.14]# ./configure && make && make install
```

The `conntrackd` daemon supports two working modes:

- **State table synchronisation:** The `conntrackd` service can be used to synchronise the packet state between several cluster members. This can be used to deploy highly available stateful firewalls. This is the main feature of the daemon.
- **Flow-based statistics collection:** In this mode the daemon is used to collect real-time statistics.

Configuring the daemon

In synchronisation mode, `conntrackd` supports up to three replication approaches:

- **notrack:** This is most simple and widely chosen, since it directly sends and receives the packet state information without doing any particular checks. This communication happens through the use of UDP.
- **ft-fw:** This mode is based on a reliable protocol that keeps performing message tracking. It would be useful if you have a very congested sync connection. This protocol can recover from message loss, re-ordering and corruption.
- **alarm:** This method can be called as spamming or spontaneous. It is an alarm-based protocol, which periodically keeps sending the flow state to the backup cluster member. The only disadvantage with this protocol is that it takes a lot of bandwidth. However, it resolves synchronisation problems rapidly. Opt for this method only when you have a dedicated link configured between cluster members.

To configure `conntrackd` in any of the above synchronisation modes, you have to copy the example configuration file to the directory `/etc/conntrackd/` on every cluster member. If `/etc/conntrackd/` is not already created, create the directory, and then copy the configuration file:

```
[root@firewall1 conntrackd]# cp /opt/hacluster/conntrack-tools-0.9.14/doc/
sync/ftfw/conntrackd.conf /etc/conntrackd/
```

Do not forget to edit the files before going ahead. There are several parameters that you have to tune to adapt the example configuration file to your set-up.



Note: If you don't want to put the configuration file under `/etc/conntrackd/`, just tell `conntrackd` where to find it by passing the option `-C`.

Now let's configure `conntrackd.conf` and `keepalived.conf`, which we had moved into the `/etc/conntrackd/` and `/etc/keepalived/` directories, respectively. For this scenario, I have used the *ft-fw* mode. Therefore, copy the `conntrack.conf` file from `$PATH/conntrack-tools-0.9.14/doc/sync/ftfw/` (where `$PATH` is a source directory). In the same way, copy `keepalived.conf` from `$PATH/conntrack-tools-0.9.14/doc/sync/` to `/etc/keepalived` and `$PATH/conntrack-tools-0.9.14/`